Desert, Forest, Ice

The world's land in 1900 and now, from HYDE 3.3

Tristan Hotham

Independent researcher
ORCID: 0000-0003-4931-9349
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Abstract

I compare the use of the world's land in 1900 and 2023 using the gridded estimates of the History Database of the Global Environment, version 3.3 (HYDE), at 5 arc-minutes, with HYDE's anthromes and its map of the potential forest zone, and check the present against MODIS satellite land cover for 2019. In HYDE's baseline estimate, cropland grew from 9.0 to 16.3 million km^2 and grazing land from 15.7 to 32.0 million km^2, so that farmland rose from 19.1% to 37.3% of the ice-free land outside Antarctica. The largest gains were in sub-Saharan Africa and South America, and by country in China, Brazil and Australia; Europe is the only region with less farmland than in 1900. Within the potential forest zone the farmed share rose from 16.4% to 27.3%, a net 6.4 million km^2, the result of 7.6 million km^2 going into farming, mostly in the tropics, and 1.2 million km^2 going out of it, 87% of that in Europe, Russia and the United States. The anthromes show that wildlands barely changed in extent (25.2% to 24.0% of ice-free land) while seminatural lands shrank from 40.1% to 24.1%. I discuss what a modelled reconstruction of 1900 can and cannot show.

Keywords: land use; land cover; HYDE; anthromes; deforestation; reforestation; farmland; historical geography; maps

1 Introduction

The twentieth century remade the surface of the land. Railways, steamships, refrigeration and later tractors, fertiliser and roads into the tropics brought new land into farming on every continent, while in parts of Europe and North America farms were abandoned and forests grew back (Williams, 2003; Ramankutty and Foley, 1999). Several groups have reconstructed this history on global grids: the SAGE cropland and pasture maps (Ramankutty and Foley, 1999), the Land Use Harmonization dataset used by climate models (Hurtt et al., 2020), and HYDE (Klein Goldewijk et al., 2017), which reaches back to 10,000 BCE and is updated regularly. Their maps are used mostly as model inputs; they are rarely looked at.

This study takes the latest HYDE release, 3.3 (Klein Goldewijk et al., 2023), and asks simple questions of two of its time slices, 1900 and 2023. How much of the world's ice-free land was farmed, and how much is now? Where did the change happen? How much of the land where forest would naturally grow is farmed, and where has farmland retreated? How have the anthromes, HYDE's classes of land by people and use, shifted between wild, seminatural and used? I present the answers as an interactive swipe map (https://maps.tristanhotham.com/land-cover-1900/) and in this short descriptive paper. It describes a reconstruction, not observations of 1900, and Section 5 sets out what that means.

2 Data

HYDE 3.3. The History Database of the Global Environment (Klein Goldewijk et al., 2023, 2017) estimates land use and population on a global grid of 5 arc-minutes (4320 by 2160 cells; about 9 km at the equator), excluding Antarctica. I use the baseline scenario (`hyde33_c7_base_mrt2023`) for 1900 and 2023: cropland, pasture, rangeland and converted rangeland in km^2 per cell, the anthromes for both years, and the general files giving each cell's maximum land area, its country and its potential forest status. HYDE allocates national and subnational totals of cropland and grazing land to the grid using weighting maps of population, suitability and terrain; before FAO statistics begin in 1961 the totals come from historical sources and from population and land per person. Grazing land is divided into pasture (more intensively managed, in wetter areas), rangeland (less intensively used, mostly in drier areas) and converted rangeland, rangeland in the forest zone that is assumed to have been cleared. HYDE 3.3 is published under CC BY 4.0.

Anthromes. HYDE provides anthromes (Ellis et al., 2010, 2021) for each time slice: 20 classes of land from urban and mixed settlements, through villages, croplands and rangelands, to seminatural woodlands and drylands and wild woodlands, drylands and ice. The split between woodlands and drylands follows potential natural vegetation.

Potential forest zone. HYDE's general file `forest_wwf_cr` marks the cells whose potential natural vegetation is forest, from the WWF terrestrial ecoregions (Olson et al., 2001). I call these cells the forest zone.

MODIS. For a present-day check I use the MODIS MCD12C1 version 6 land cover type product for 2019 (Friedl and Sulla-Menashe, 2019), the majority IGBP class on a 0.05 degree grid (public domain).

HYDE 3.2 world series. For the long series in Figure 2, I use the world totals of cropland and grazing land from HYDE 3.2 for 1700 to 2016 as republished by Our World in Data (Our World in Data, 2017), since the HYDE 3.3 time slices I used are only 1900 and 2023. The two versions differ: HYDE 3.2 gives 8.4 million km^2 of cropland in 1900 against 9.0 million in 3.3.

Geography. Country names and outlines come from Natural Earth (Natural Earth, 2026); HYDE's numeric country codes are matched to them by ISO number.

3 Methods

Farmland. Farmland is cropland plus grazing land, where grazing land is pasture, rangeland and converted rangeland (HYDE's own grazing total equals this sum to within rounding, and converted rangeland lies only in the forest zone). Shares are of HYDE's maximum land area. Ice-free land excludes land in cells of the anthrome 'ice, uninhabited' in 2023 (2.6 million km^2, almost all in Greenland and the Arctic islands), leaving 129.2 million km^2.

Forest zone. In each forest-zone cell, the farmed area is cropland plus pasture plus converted rangeland, and the unfarmed area is the rest of the land. This is a measure of farming in the forest zone, not of forest: logging, fire, plantations and natural openings are invisible to it. Gross gains and losses are sums of cell-by-cell increases and decreases between 1900 and 2023; the net change is their difference.

Countries, regions and hotspots. I summed cells by HYDE's country grid and grouped countries into twelve regions using Natural Earth subregions, with Russia on its own. I also defined eleven boxes for named regions (Table 2), restricted to one country where the name implies it (for ex-

ample Brazil for the Cerrado box). The boxes are approximate and meant for comparison, not as ecological boundaries.

Comparison with MODIS. I aggregated both datasets to a quarter-degree grid (3 by 3 HYDE cells; 5 by 5 MODIS cells), computed for each the share of forest-zone land that HYDE has farmed in 2023 and the share that MODIS classifies as neither forest (IGBP classes 1 to 5) nor woody savanna (class 8), and correlated the two across cells with at least 50 km^2 of forest zone. MODIS cropland is class 12 plus half of the cropland and natural vegetation mosaic (class 14).

Maps. The figures use the native 5 arc-minute grid in the Equal Earth projection; the interactive map sums the grid to 10 arc-minutes. All numbers in this paper are computed by the scripts in the repository and stored in `data/results.json`.

4 Results

4.1 Farmland

HYDE estimates 9.0 million km^2 of cropland in 1900 and 16.3 million in 2023 (×1.81), and 15.7 and 32.0 million km^2 of grazing land (×2.04). Farmland covered 19.1% of the ice-free land in 1900 and 37.3% in 2023 (Figure 1): a net gain of 23.5 million km^2, the result of 25.0 million km^2 of cell-level gains and 1.5 million of losses. Rangeland is the largest single component in both years (8.7% and 16.6% of ice-free land), followed by cropland (7.0% and 12.6%), pasture (2.3% and 5.8%) and converted rangeland (1.2% and 2.4%).

In the longer HYDE 3.2 series (Figure 2), world farmland grew by 15.4 million km^2 between 1700 and 1900, by 13.3 million in the next fifty years, by 9.5 million from 1950 to 1990, and by only 0.8 million from 1990 to 2016.

4.2 Where the change happened

Every region except Europe gained farmland (Figure 3, Table 1). Sub-Saharan Africa gained the most (5.7 million km^2), then South America (4.7), East Asia and Oceania; Europe lost 0.4 million km^2. Of the 179 countries and territories with at least 1,000 km^2 of land, farmland shrank in 26. The largest national gains were in China (2.4 million km^2), Brazil (2.4), Australia (2.4), the United States (1.1), Argentina and Kazakhstan; the largest losses in France, Poland, Italy and Germany. In Australia and China most of the gain was grazing land; in Brazil 1.1 million km^2 of the 2.4 million gained was in the forest zone.

The change map (Figure 4) shows the frontiers. The Brazilian Cerrado box went from 3.4% farmed to 40.6%, the Sahel from 26.5% to 63.5%, Southeast Asia from 12.0% to 37.1% (mostly cropland), and the Australian wheat belt from 11.4% to 51.2%. Older frontiers changed less: the US Midwest box was 50.2% farmed in 1900 and 50.4% in 2023, though its cropland share fell from 44.2% to 37.2%. Western and central Europe fell from 60.0% to 47.4%, and the eastern United States from 39.4% to 33.3%.

4.3 The forest zone

The forest zone covers 58.8 million km^2, 45.5% of the ice-free land. Its farmed share rose from 16.4% in 1900 to 27.3% in 2023 (Figure 5). The net gain, 6.4 million km^2, is 27% of the world's net gain in farmland; the rest of the new farmland is outside the forest zone, in grasslands, savannas

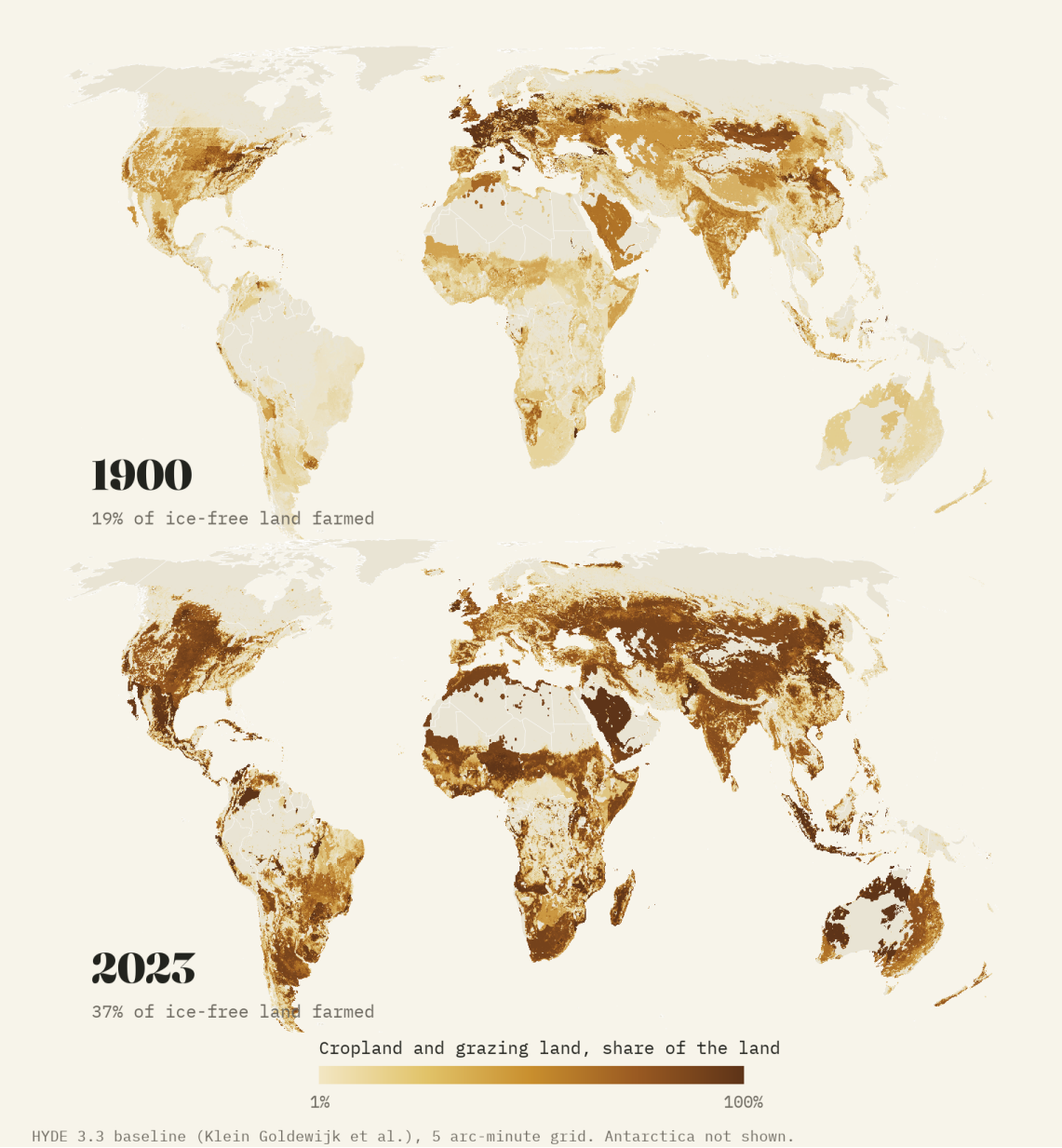


Figure 1: Farmland (cropland plus grazing land) as a share of the land in each 5 arc-minute cell, 1900 and 2023. HYDE 3.3 baseline.

Table 1: Farmland by region, 1900 and 2023: ice-free land (million km^2), farmland (million km^2), farmland as a share of ice-free land (%), and the farmed share of the forest zone (%). HYDE 3.3 baseline.

Region	Land	Farm 1900	Farm 2023	% 1900	% 2023	Forest zone 1900	2023
Sub-Saharan Africa	21.5	3.69	9.41	17	44	13	36
South America	17.5	1.28	5.95	7	34	4	22
East Asia	11.5	4.23	6.71	37	58	30	46
Oceania	8.5	1.04	3.46	12	41	7	31
North Africa and West Asia	12.8	2.30	4.38	18	34	36	63
North America	17.7	3.02	4.60	17	26	12	13
Central Asia	3.9	1.72	2.92	44	75	21	37
South Asia	6.6	2.00	3.18	30	48	36	55
Southeast Asia	4.5	0.45	1.61	10	36	10	36
Central America and Caribbean	2.7	0.52	1.44	20	54	13	39
Russia	16.3	1.65	2.16	10	13	8	8
Europe	5.8	2.82	2.44	49	42	50	42

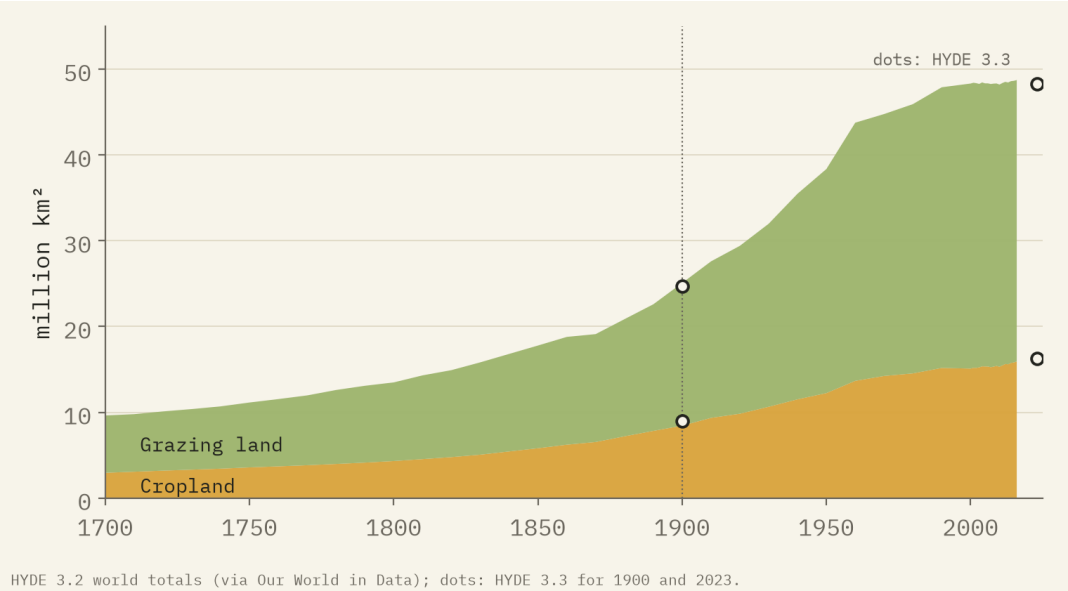


Figure 2: World cropland and grazing land, 1700 to 2016, HYDE 3.2 via Our World in Data. Dots: HYDE 3.3 for 1900 and 2023.

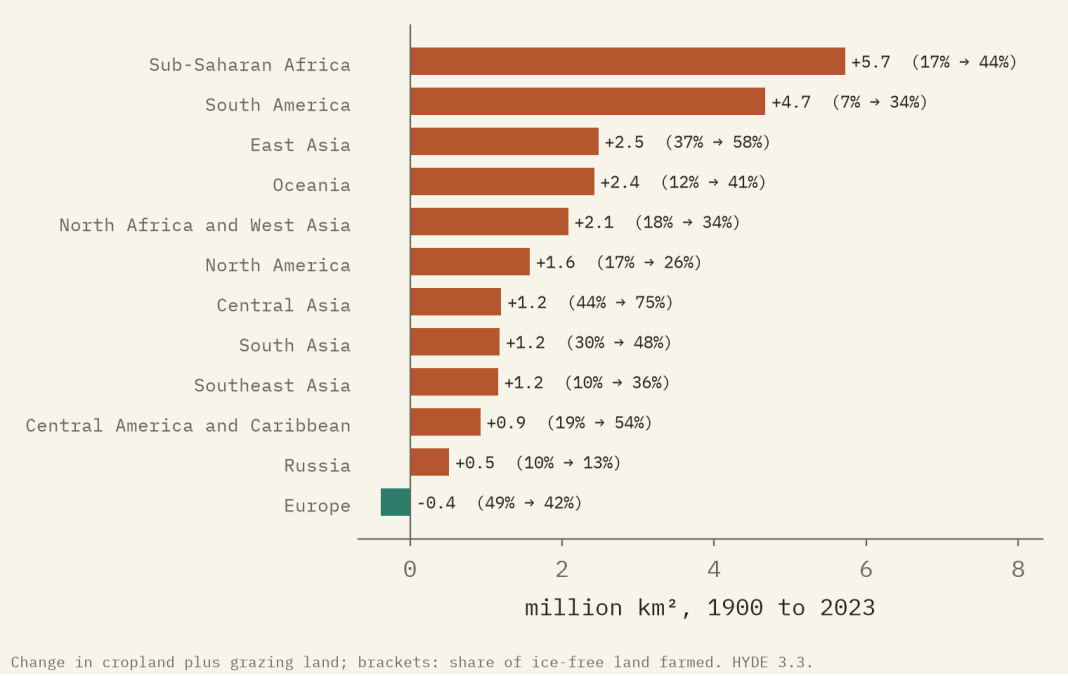


Figure 3: Change in farmland by region, 1900 to 2023, with the share of ice-free land farmed in each year.

Table 2: Named boxes: latitude and longitude limits, and farmland and cropland as a share of their land (%), 1900 and 2023. Boxes named after a country are limited to that country.

Region	Latitude	Longitude	Farm 1900	Farm 2023	Crop 1900	Crop 2023
Brazilian Cerrado	-24 to -2	-60 to -41	3	41	0	9
Southern Amazon arc	-15 to -2	-70 to -44	1	23	0	4
Argentine Pampas	-40 to -27	-66 to -56	18	68	3	33
US Great Plains	30 to 49	-105 to -95	46	80	18	33
US Midwest and Corn Belt	36 to 49	-95 to -80	50	50	44	37
Kazakh steppe	46 to 55	60 to 80	50	82	13	16
Sahel	10 to 17	-17 to 38	26	64	6	22
Southeast Asia	-11 to 24	92 to 141	12	37	9	32
Australian wheat belt	-36 to -26	113 to 125	11	51	0	10
Eastern United States	24 to 50	-90 to -66	39	33	35	24
Western and central Europe	36 to 60	-10 to 25	60	47	37	31

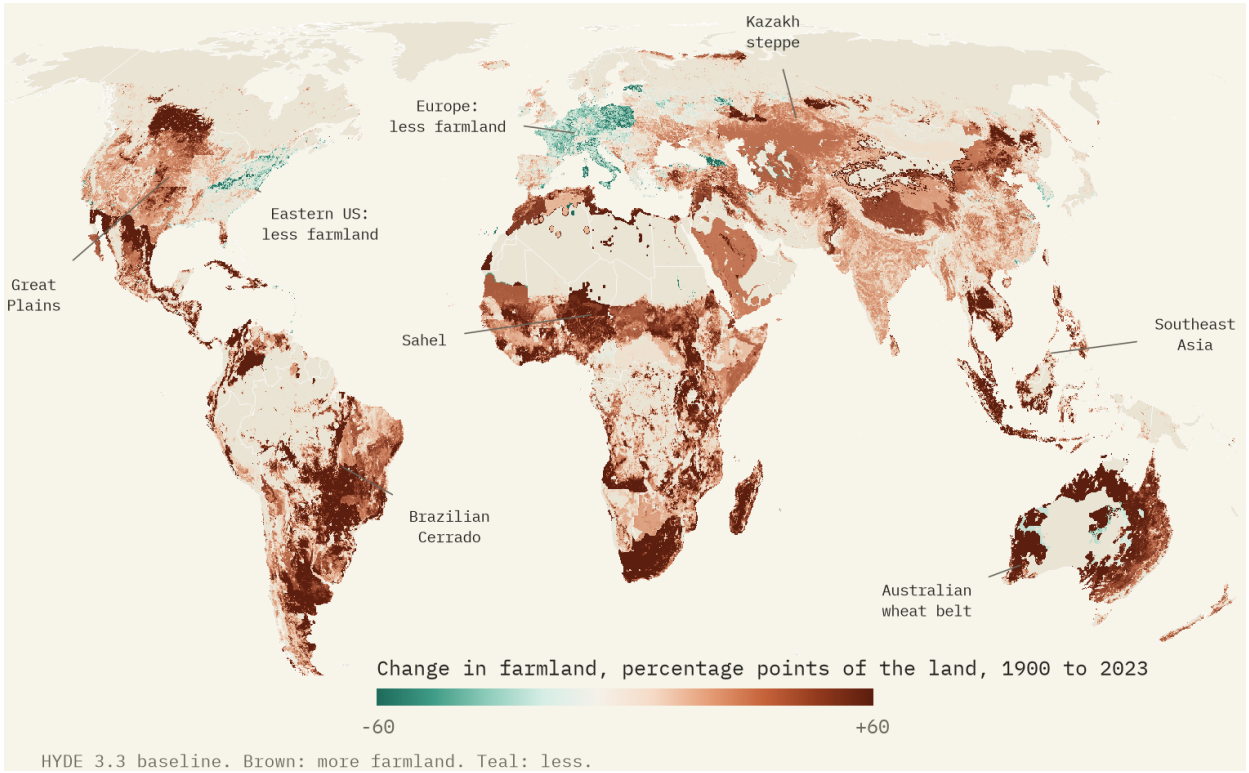


Figure 4: Change in farmland share of the land, percentage points, 1900 to 2023.

and drylands. Gross gains of 7.6 million km^2 were led by Brazil (1.1 million), China (0.8), Indonesia (0.5), Australia and India. Gross losses of 1.2 million km^2 were concentrated in Europe, Russia and the United States (87%); France, Poland, Italy and Germany had the largest net declines. In the United States east of 90°W the farmed share of the forest zone fell from 38.3% to 31.2%, and this area holds 90% of the country's gross decline. In Europe it fell from 50.1% to 41.7%.

4.4 Anthromes

Used anthromes (dense settlements, villages, croplands and rangelands) covered 34.6% of ice-free land in 1900 and 51.9% in 2023 (Figure 6). Wildlands changed little (25.2% to 24.0%); 94.0% of the land that was wild in 1900 is still wild, and 5.1% became rangeland. Wild woodlands shrank by 2.7%, and in 2023 77% of them lie north of 50°N. The large change was in seminatural lands, which fell from 40.1% to 24.1%, while villages rose from 2.5% to 7.7% and dense settlements from

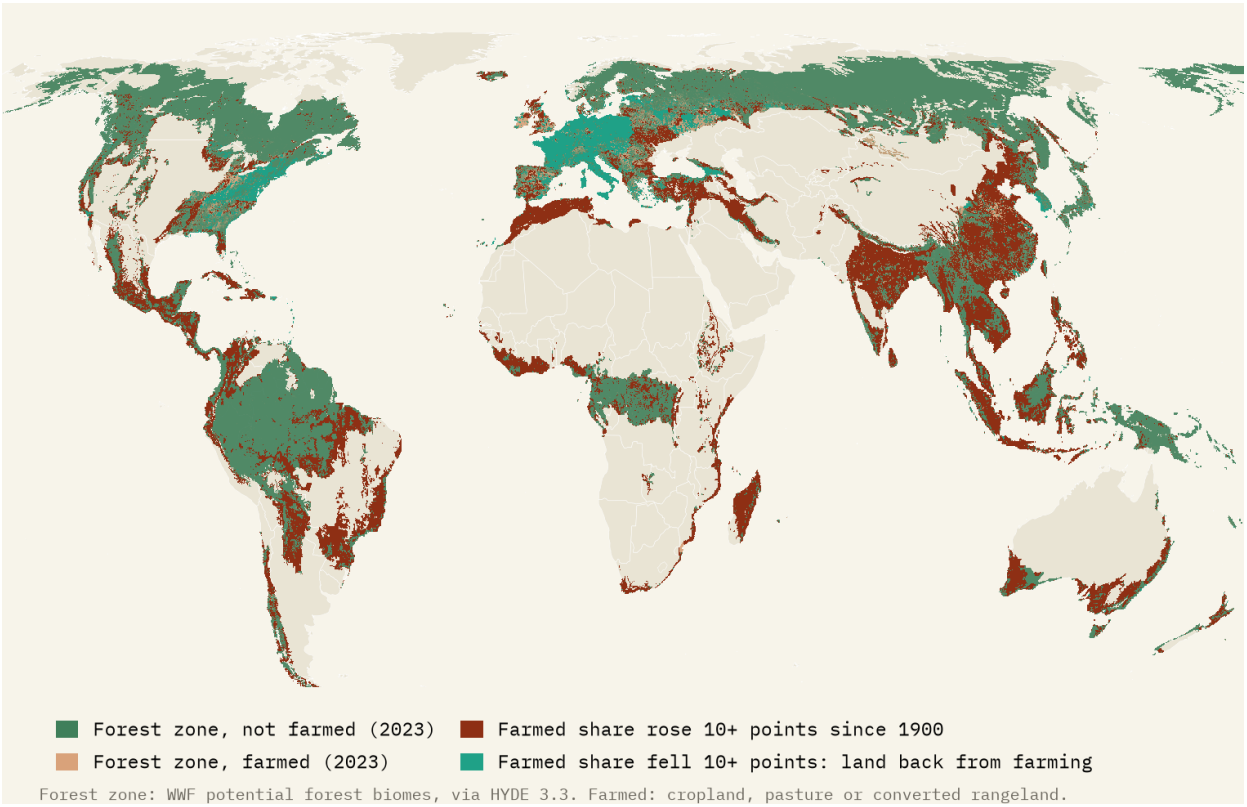


Figure 5: The potential forest zone in 2023: unfarmed and farmed land, and cells where the farmed share rose or fell by at least 10 percentage points since 1900.

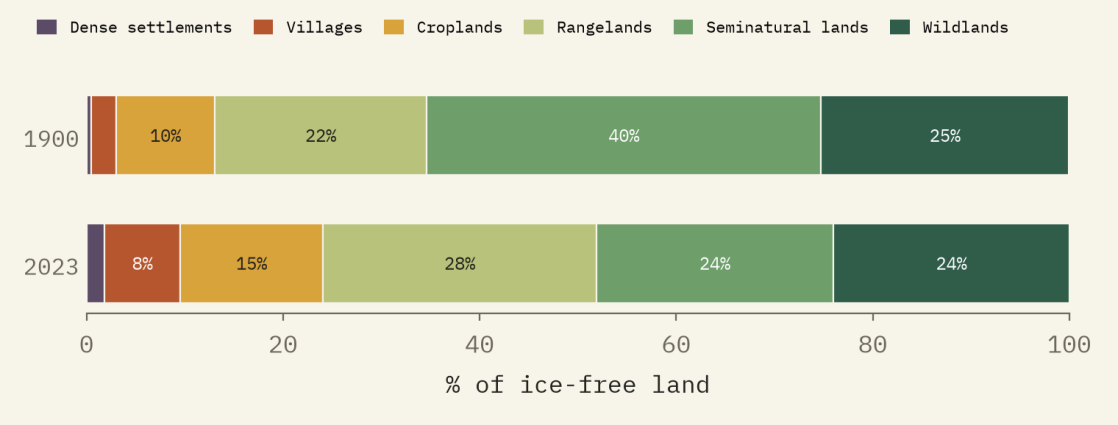


Figure 6: Anthrome groups as a share of ice-free land, 1900 and 2023. HYDE 3.3.

0.5% to 1.8%.

4.5 Comparison with satellite land cover

MODIS classifies 22.5 million km^2 as forest in 2019 and 11.4 million as woody savanna, 20.0 million as barren and 2.1 million as snow and ice on HYDE's land (Figure 7). Within the forest zone, MODIS finds forest on 35.7% and forest or woody savanna on 52.7%, against 72.7% that HYDE has unfarmed. Across 120,913 quarter-degree cells, the correlation between HYDE's farmed share and the MODIS share without forest or woody savanna is 0.49. MODIS cropland (13.1 million km^2) is 80% of HYDE's 2023 cropland. The two datasets measure different things: HYDE land use from statistics, MODIS land cover from reflectance, and the gap between unfarmed and forested land

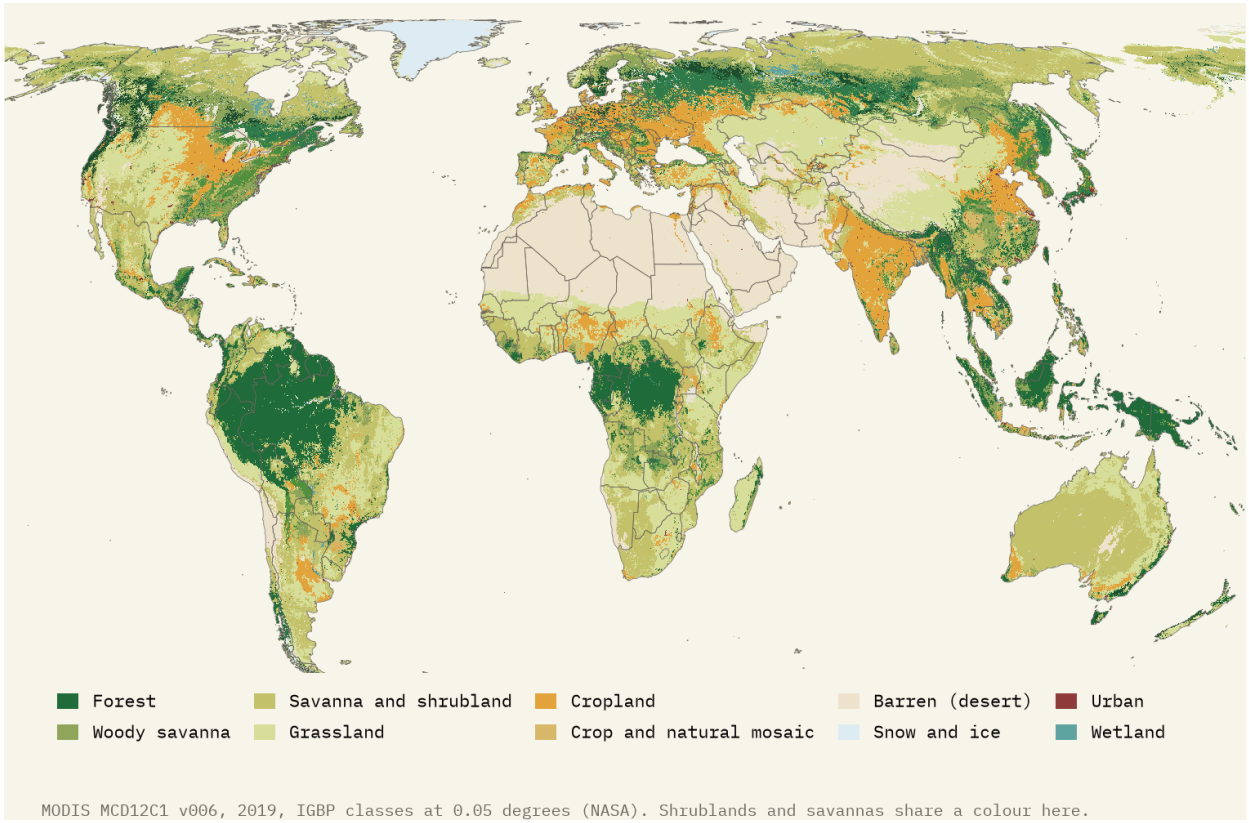


Figure 7: Land cover in 2019 from MODIS MCD12C1 (IGBP classes, grouped for display).

is degraded, logged, burnt or naturally open land that HYDE does not distinguish.

5 Limitations

First, the 1900 map is a reconstruction. HYDE's totals for 1900 rest on historical statistics of variable quality and, where these are lacking, on population and assumed land per person; their spatial pattern comes from allocation rules. Country and regional totals are more reliable than individual cells, and uncertainty is largest where statistics are weakest, in much of Africa, Asia and Latin America. HYDE 3.3 also offers lower and upper estimates, not used here. Second, grazing land is a broad category: most of it is rangeland, which may look natural, and the boundary between grazed and ungrazed land is uncertain. Third, the forest zone is potential vegetation, and farming is the only change HYDE records in it; it does not measure forest cover, logging, fire, plantations or natural regrowth on unfarmed land. Fourth, the long time series uses HYDE 3.2, which differs in detail from 3.3. Fifth, MODIS is itself a classification with errors, at a different date and with different definitions, so the comparison tests consistency rather than accuracy. Finally, the regions and hotspot boxes are my choices.

6 Conclusion

In HYDE's reconstruction the share of the world's ice-free land used for crops and grazing roughly doubled between 1900 and 2023, from about a fifth to more than a third. The expansion moved from the temperate frontiers of the nineteenth century into the tropics, the steppes and the drylands; Europe and the eastern United States gave land back. The wild that remains is mostly the

land that was always hard to farm. Set side by side on a swipe map, the two years make the scale and geography of that change easy to see, while the data's limits remain visible.

Data and code availability

The interactive map is at https://maps.tristanhotham.com/land-cover-1900/. Data and code are at https://github.com/Tristan-Hotham/land-cover-1900 and archived with this paper at https://doi.org/10.5281/zenodo.23847971. Derived data are released under CC BY 4.0 and code under the MIT licence. HYDE 3.3 is CC BY 4.0 (DOI 10.24416/UU01-94FNH0); MODIS data are public domain; Natural Earth is public domain.

AI statement

I wrote the text, with drafting and editing assistance from Claude (Anthropic), which also helped prepare the analysis code. I reviewed and checked all text and results.

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